Recommendation Systems for Personalized Content Discovery

Netflix Prize Dataset · SVD vs ALS · End-to-End Full-Stack Deployment

Body:

- 100M+ ratings · 480K users · 17,770 movies
- Two models compared: Matrix Factorization (SVD) and Implicit Feedback ALS
- Evaluated on RMSE and MAP@10 · Deployed as a React + Node.js web app

The Recommendation Problem

The Challenge:

- Users face thousands of titles; attention is finite
- Raw ratings data is 95.8% sparse — most user-movie pairs are unobserved
- Long-tail distribution: top 5% of users generate ~30% of all interactions
- Cold-start: new/sparse users receive the worst recommendations

Our Approach:

- Density-preserving subsample: 10K users × 5K movies
- Two distinct algorithmic paradigms compared head-to-head
- Evaluation on both rating accuracy AND ranking quality
- Explainability layer + cold-start mitigation strategies

Data-Driven Decisions Start with Honest Exploration

Top row:

- ~10,000 Users
- ~5,000 Movies
- ~2.1M Interactions
- 95.8% Sparse

Main insight:

- **Positivity bias** — ratings of 3–4 dominate; 1-star is rare. Users self-select content they expect to enjoy.
- **Power law activity** — top 5% of users generate 30% of ratings. Models must handle the other 95%.
- **61.4%** of ratings qualify as relevant (≥3.5★) — the MAP@10 threshold is non-trivial and well-calibrated.
- **Temporal signal** — rating activity peaks 2004–2005, monthly seasonality visible — future models can leverage recency weighting.

Two Paradigms, One Question: What Should You Watch Next?

SVD (Explicit Feedback)

$R \approx P \times Q^T$ + biases

Optimized by: SGD on observed ratings

Learns: "What would you rate this?"

Best metric: RMSE

- 100 latent factors
- 30 epochs, lr=0.005, reg=0.02
- Captures user/item bias terms

ALS (Implicit Feedback)

$C_{ui} = 1 + 40 \times r_{ui}$

Optimized by: closed-form least squares

Learns: "How likely are you to watch this?"

Best metric: MAP@10

- 100 factors, 30 iterations
- Treats unobserved items as weak negatives
- Parallelisable — 33% faster than SVD

Key difference:

SVD ignores what users *didn't* rate. ALS learns from it. This is why ALS ranks better.

What the Data Taught Us

RMSE ≠ Good Recommendations SVD achieves 21% lower RMSE than ALS, yet ALS delivers 22% better ranking quality. Optimizing the wrong metric ships the wrong product.

Absence is Evidence ALS's MAP@10 advantage comes entirely from treating unobserved ratings as weak negatives. The data you *don't* have is as informative as the data you do.

Cold Users Are Underserved Users with ≤10 training ratings achieve MAP@10 of 0.041 vs 0.209 for warm users — a 79% relative gap. This is where retention risk lives.

Models Disagree More Than They Agree Only 18% average overlap between SVD and ALS Top-10 lists. An ensemble harvesting both would outperform either model alone.

The Threshold Is a Business Decision Moving the relevance threshold from 3.5 to 4.0 changes the "relevant" pool from 61% to 38% of ratings — a choice that should be validated against real engagement data, not chosen arbitrarily.

Results: SVD vs ALS

	SVD	**ALS**	**Winner**
RMSE ↓	**0.9124**	1.1247	SVD ✓
MAE ↓	**0.7156**	0.8934	SVD ✓
MAP@10 ↑	0.1832	**0.2241**	ALS ✓
Train Time	47.3s	**31.8s**	ALS ✓

Cold users (≤10 ratings): MAP@10 = 0.041

Warm users (>10 ratings): MAP@10 = 0.209

Gap = −0.168 MAP@10 points (79% relative degradation)

For a streaming platform, deploy ALS for ranked carousels. Use SVD when you need to display a star rating. Consider an ensemble for best-of-both.

The Model in Action

 # Success Case (User 2847, 312 ratings)

Rank	Title	Predicted	Actual	✓
1	Lord of the Rings	4.71★	5.0★	✓
2	Shawshank Redemption	4.68★	5.0★	✓
3	Schindler's List	4.62★	4.5★	✓

AP@10 = 0.82 — Model correctly identifies prestige drama preference cluster.

Explainability: *"Because you highly rated Shawshank Redemption (similarity=0.91), we recommend The Green Mile."*

 # Failure Case (User 6103, 8 ratings)

Rank	Title	Predicted	Actual	✓
1	Titanic	4.21★	2.0★	✗
2	Forrest Gump	4.18★	3.0★	✗
3	The Notebook	4.15★	2.5★	✗

AP@10 = 0.00 — Popularity bias dominates; user's real preferences are invisible with 8 interactions.

Item-similarity bridging from their 8 seed ratings.